

Does Keystroke Dynamics tell us about Emotions?

A Systematic Literature Review and Dataset Construction

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Abstract—There is strong evidence that emotional states affect the Human’s performance and decision-making. Therefore, understanding Human emotions has become of great concern in the field of Human-Computer Interaction (HCI). One way to an online emotion recognition is through keystroke dynamics. Keystroke Dynamics focuses on the particular way a person types on a keyboard. To provide insight, assess recent works, and guide future researches in this field, a Systematic Literature Review (SLR) is conducted. An SLR adopts a rigorous procedure with the definition of a formal review protocol.

The primary aim of this paper is to highlight the effectiveness of using keystroke dynamics biometrics in recognizing emotions by systematically presenting research efforts in the past decade. We provide insight into such an approach by summarizing and discussing the data acquisition procedures, datasets, extracted features, classification methods, and performance measures used in previous researches.

In light of these findings and the noticed scarcity of datasets related to emotion recognition through keystroke dynamics, we develop and host an interactive web application to construct a new dataset. The findings of this work reveal new interesting research directions and may motivate the research community to develop automatic emotion recognition systems based on keystroke dynamics.

Index Terms—HCI, keystroke dynamics, emotion recognition, data collection, web application, Systematic Literature Review (SLR).

I. INTRODUCTION

Emotion is a fundamental component of a Human being. There is strong evidence that the different emotional states motivate action and add meaning and richness to virtually all human experiences [1]. Because of the widely increasing number of digital gadgets that are being used, developing intelligent systems that are capable of understanding Human emotions has become the aim of many researchers and practitioners. In psychology, the patterns of emotions are extensively and widely studied. However, we cannot find an exact definition of the term. One definition given by Plutchik [2] is: “*An emotion is not simply a feeling state. Emotion is a complex chain of*

loosely connected events that begins with a stimulus and includes feelings, psychological changes, impulses to action and specific, goal-directed behavior”. What makes emotions distinguishable from moods and personal traits, is that they are short-lived, intense and can last from seconds to hours [3]. To explain the nature of emotions, psychologists developed 3 major emotion models: (i) Categorical model in which emotions are grouped into discrete categories. This model was introduced by Ekman [4] as the six basic emotions; (ii) dimensional model which define regions within which distinct emotions can be located; two, three and four-dimensional models are distinguished [5]; (iii) cognitive appraisal model, known as the OCC (Ortony, Clore, and Collins) theory [6], views emotions as elicited by specific types of appraisals—positive or negative judgments—made by an individual.

Studies based on neurology have proved the correlation between affective states and physical behavior such as hand motions [7]. Emotions can induce strong effects that may go from gross changes in our internal environment to small fluctuations in our muscles [8], i.e. the rhythm, with which human fingers are used to type on a keyboard, can be different for different emotional states. In light of these findings, research on keystroke dynamics biometrics applied to emotion recognition has increased showing promising results. Such a practice has gained momentum because when compared to other biometric systems, keystroke dynamics are quite easy to implement in real-life scenarios with minimal alteration and user intervention, and most importantly, it is not expensive at all.

The present paper provides some guidance to researchers and practitioners by exposing current approaches and making connections among the diverse areas of knowledge about emotion recognition through keystroke dynamics. For this purpose, we adopted a Systematic Literature Review (SLR) as defined in [9]. According to Kitchenham and Charters’ guidelines, SLR are rigorous, auditable and are generally based on three main phases namely; planning, conducting, and reporting as described in Fig. 1. In the first phase, the objective

of the review is illustrated, the research questions are defined and a clear review protocol is developed. The second phase concerns conducting the review, in which the review protocol is applied and primary research is selected and then assessed. As for the third phase, it encompasses answers for research questions previously specified.

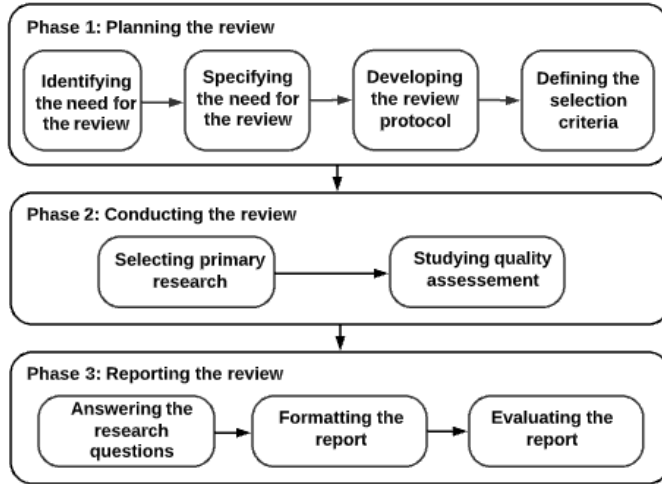


Fig. 1: Steps for a Systematic review according to Kitchenham

The remaining part of the paper is organized as follows: Section II introduces the research questions and describes the adopted SLR protocol. Section III explains the conduction of the review while applying the developed protocol. In Section IV, the results obtained by the SLR are summarized and discussed when answering the research questions. Finally, a further research proposal and a conclusion are given in sections V and VI, respectively.

II. PLANNING THE REVIEW

In a very first step, we searched for systematic reviews that hold the same goals as the present work. No publications were detected. Thus, the need for conducting such a review was identified.

A. Specifying the research questions

In light of the need for a systematic review, the present paper addresses the following research questions:

- *RQ1: What are the trends in online emotion recognition technologies?*
- *RQ2: Keystroke dynamics in emotion recognition; what applications, strengths, and weaknesses?*
- *RQ3: What are the different features extracted from a user's particular typing rhythm?*
- *RQ4: Which classification techniques used for emotion recognition through keystroke dynamics?*
- *RQ5: What are the measures used for evaluation and what is the performance achieved?*
- *RQ6: What about the existing datasets related to emotion recognition through keystroke dynamics?*

B. Developing the review protocol

In this phase, a search strategy was set and applied. We started by identifying the terms related to the research questions: {keystroke dynamics, typing dynamics, keystroke biometric, keystroke pattern, typing pattern, emotion detection, emotion recognition, emotion-sensing, affective computing}. After that, a research string using the above terms was built using the logical connectives "OR", "AND", and "AND NOT". The search process was performed in the indexed and largest electronic database SCOPUS (www.scopus.com). Fig. 2 presents the search string applied.

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TITLE-ABS-KEY (((("emotion detection" OR "emotion
recognition" OR "emotion sensing" OR "affective
computing") AND ("keystroke dynamics" OR "typing
dynamics" OR "keystroke biometric" OR "keystroke
pattern" OR "typing pattern")) AND NOT ("review"
OR "systematic review")))

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Fig. 2: Research string used for paper extraction

We should note that *RQ1* was studied at a very early stage i.e. separate from the remaining questions. That's why we defined another research string to answer *RQ1*. We used the following terms to be applied in the search string: {emotion recognition, emotion detection, technologies, methods, approaches, online, automated, real-time}.

C. Selection criteria

In order to obtain primary studies that provide direct evidence about the previously defined research questions, a set of selection criteria is required. Therefore, inclusion and exclusion criteria were applied to the returned references.

Only journal and conference papers that are written in English and published after the year 2011 were included in this study. However, we excluded short papers (less than 4 pages), papers that are not available online, incomplete studies, or those that do not present any results. For the returned papers to be directly aligned with the research questions, we ignored multimodal approaches.

As for the search process that we conducted separately for the first research question, we used the same criteria except that we limited the search to publications after 2018. The reason behind that is that there is a large number of publications. Moreover, the returned references do not make the main interest of the present work.

III. CONDUCTING THE REVIEW

The conduction phase was performed based on the search strategy defined in the planning phase. As mentioned in the previous section, the search process for *RQ1* and the remaining *RQs* were conducted separately.

A. Selecting primary research

After applying the search string, we firstly obtained a sum of 716 papers. 692 of them related only to *RQ1*. After

applying the inclusion and exclusion criteria, the number was reduced to 29 (19 papers related to *RQ1* and 10 papers related to *RQ2* to *RQ6*).

B. Studying quality assessment

The number of papers retained was not very big; hence, the quality assessment process consisted in assessing each paper separately to make sure that it satisfies the research questions. This was assured by reading parts of articles like abstracts, keywords, and conclusions.

IV. REPORTING THE REVIEW

We report here the answers to the research questions.

A. *RQ1: What are the trends in online emotion recognition technologies?*

With regards to the research we conducted, we found that, since 2018, 19 publications dealt with the different technologies of emotion recognition. All these papers reveal that emotions can be sensed from: facial expressions, body gestures and movements, physiological states, speech and voices, text, and keyboard and mouse dynamics. Table I reports a summary of the different technologies applied.

While investigating the number of papers published in the two previous years, we notice that the majority of technologies to detect emotions use physiological states and facial expressions. Such techniques reveal significant accuracy; however, they demand additional expensive equipment and hardware setup. As for detecting emotions based on physiological states, equipment used may have intrusive nature; sensors take time to attach and shaving may be necessary. All these approaches require the use of additional devices that, other than being not realistic in a casual usage context, they can affect the natural emotional state of a user. In textual emotion recognition, obstacles like spelling errors, languages and slang can be faced. Additional steps must be performed and an ontology, as well as a semantic analysis, is required.

B. *RQ2: Keystroke dynamics in emotion recognition; what applications, strengths, and weaknesses?*

Keystroke dynamics refer to the process of measuring and assessing a user's typing activities such as rhythm pattern, overall speed, common errors, and others on digital devices. These devices may refer to a computer keyboard, mobile phone, or touch screen panel. The keyboard is a device that can provide useful information about the user, thus exploited in many areas such as: emotion detection [28]–[37], Identification and authentication [31], [38], and gender estimation [39].

The diversity of keystroke dynamics applications has given rise to more research efforts and thus, to a growing number of publications during the last decade. Fig. 3 shows the evolution of studies related to emotion recognition through keystroke dynamics. As mentioned in

section II-C, reviews and papers not presenting a new methodology were not taken into consideration.

The evolution of the number of studies depicted by the shape of the curve in Fig. 3 can be explained by the various advantages that keystroke dynamics offer, unlike other technologies. The most strengths to mention are: transparency and non-intrusiveness, low implementation and deployment cost, ease of integration into the computer with minimal user intervention, prompt recognition process (real-time), and adaptability in remote devices. However, some weaknesses can be faced; e.g., the big amounts of data should be gathered for the training process, the instability along time (i.e. the values of keystroke parameters taken from a user may depend on the type of software used). Moreover, possible decrease in accuracy due to the variations in typing rhythm, caused by external factors like injury, fatigue, or distraction, may occur.

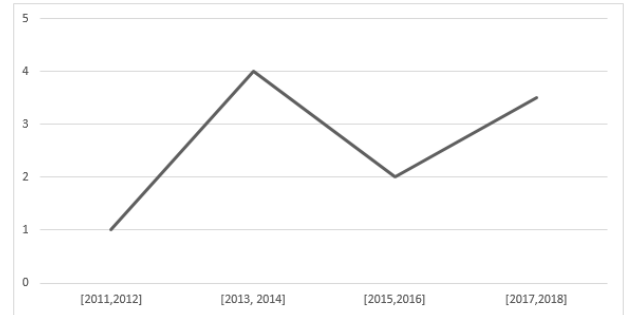


Fig. 3: Evolution of studies conducted in the field of emotion recognition through keystroke dynamics

C. *RQ3: What are the different features extracted from a user's particular typing rhythm?*

While a user is using a keyboard to type any kind of data, the following information can be recorded and stored:

- The key pressed
- The timestamp of the key press
- The timestamp of the key release

Based on the above information, features can be extracted to be used as inputs for classification or feature selection process.

In Fig. 4, we summarize the commonly used features in keystroke dynamics related studies. The majority of selected papers extracted the timing-based features including duration, latency, and typing speed. Very few studies extracted frequency related features. When measuring duration and latency, papers like [28]–[30] used di-graphs and tri-graphs. The rest of papers either did not mention the number of graphs used or used a single graph. A list of the features used in each of the retained papers is given in Table II.

In [35], features' indications on emotions were concluded; lower typing speed and higher error rates

TABLE I: Most recent emotion recognition technologies published since 2018

Technologies	Description	Equipment	Ref.
Physiological states	Use of physiological metrics such as Galvanic Skin Response (GSR), heart rates, respiration, electrocardiography (EKG), electromyography of the jaw (EMG) to detect emotional impulses changes.	Electronic sensors	[10]–[17]
Body gestures	Sensing emotions based on body language such as tapping with the foot, crossing the arms, tilting the head, changing the position a lot of times while being sat, etc.	Video camera	[18]
Facial expressions	Face tracking software are used to analyze facial expressions. Such information is gathered from webcam images to predict emotions.	Video camera	[19]–[23]
Text content	Emotion detection from texts, referred to as sentiment analysis, analyses the words contained on a message and predict the emotional state of the user based on a previously trained model.	-	[24]–[26]
Speech and voice	By analyzing the syntactic-semantic part of a speech as well as the paralinguistic information of the speaker (tone, emotional state, and gestures), the emotional state of a participant can be extracted.	Microphone	[19], [25], [27]
Keyboard and mouse dynamics	Detection of user emotional states through their typing rhythms, mouse movements, handwriting, etc.	Keyboard, mouse and/or touchscreen	[28]

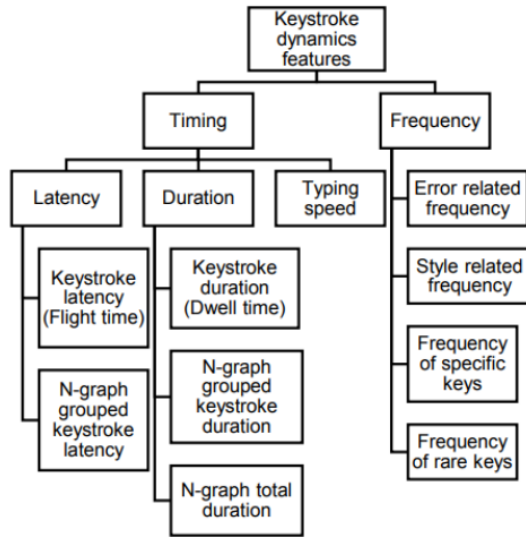


Fig. 4: Keystroke dynamics features

TABLE II: Keystroke dynamics features extracted from retained studies

Features	Description
Latency or flight time	Time interval between key release of a keystroke and key press of the following keystroke. It should be noted that this feature can also hold negative values. This happens when a key has not yet been released and a new one is pressed. [28]–[34], [36]
Duration or dwell time	Time interval between a key press and release of the same key. [28]–[34], [36], [37]
Typing speed	Total number of keystroke or words per unit of time (seconds). [35]
Frequency	Frequency of error, special keys (enter, backspace, delete, and arrow keys). [31], [35], [36]
Other features	Pressure (in touchscreens), input rate, pause rate, key code. [35], [37]

are indicators of negative emotions and high valence such as stress. In [33], the results further indicate that the keystroke duration is more sensitive to valence, whereas the accuracy rate is more sensitive to arousal.

D. RQ4: Which classification techniques used for emotion recognition through keystroke dynamics?

Commonly deployed classification techniques are mainly categorized as statistical or machine learning approaches:

- *Statistical approach:* the related methods are generally simple, easy to implement and have low overhead. Diverse statistical measures are used namely; mean, median and standard deviation, statistical t-test, and k-nearest neighbor. However, using a linear statistical approach may not be appropriate in the case of keystroke dynamics-based detection due to the non-linearity of the features. Besides, using such a technique will overlook the usefulness of the training stage in identifying patterns.
- *Machine learning approach:* consists in identifying and classifying patterns based on provided data. Some of the machine learning techniques that are most used in emotion recognition through keystroke dynamics are: nearest neighbor algorithms and clustering, data mining, Bayes classifier, Support Vector Machine, Fuzzy logic, decision tree, etc.

With regards to the methods used in emotion recognition through keystroke dynamics, we found that most of the researchers adopted machine learning rather than statistical methods. Table III outlines the different classification algorithms applied in the works undertaken by researchers towards developing an emotion recognition system. Some authors proposed new algorithms. In [31], a new light-weight classifier namely Bounded K-means Clustering was proposed.

E. RQ5: What are the measures used for evaluation and what is the performance achieved?

After conducting the process of recognition, it is essential to evaluate the techniques implemented. There are different evaluation criteria: Accuracy rate, False Negative Rate (FNR), False Positive Rate (FPR), Kappa, F-score, and Correlation analysis, etc.

Since some of the selected papers include mouse movements along with typing patterns, we investigated only the accuracy rates of keystroke dynamics. Table III reveals that the best performance results are inspected in [29] and [36]. In [29], an accuracy rate of 91.24% was achieved with 4.35% FPR using SVM for “fright” emotion. In [36], Logistic regression, SVM, Nearest neighbor, C4.5, and Random Forest were applied to the data. The best accuracy rates were obtained using KNN reaching 84% for arousal and 83% for valence. However, we should note that the dataset used in [29] is very small (4 emotions) compared to [36] (14 emotional states using the Arousal-Valence model).

Although we are dealing with a real-time emotion recognition system, the computational time i.e. the time consumed in each classification algorithm from the beginning to the end was not considered in all the selected papers.

F. RQ6: What about the existing datasets related to emotion recognition through keystroke dynamics?

It is important to note that datasets related to emotion recognition through keystroke dynamics are very scarce. That’s why, authors start by constructing the datasets. To do so, they should answer the following questions:

- How raw data will be collected?
- What is the emotion model that will be used? (Categorical, dimensional, or cognitive appraisal model as described in section I).
- Which method will be chosen for data labeling? Or an emotion induction method will be used?
- What type of text for data entry (Fixed or free text)?
- What is the temporal applicability of the model?

As depicted in Table III and Table IV, each paper deals with different parameters. Referring to the completely different parameters used, benchmarking datasets becomes difficult, hence assuming the best recognition method would not be fair. However, we notice that studies with discrete emotion models, fixed texts, and considering a higher number of participants present slightly higher accuracy rates.

V. DEVELOPING THE RESEARCH PROPOSAL

Based on the conducted SLR, scarcity of datasets, and the fact that performance in keystroke dynamics is highly dependent on the dataset, our purpose is to develop a dynamic web application to construct a new dataset. Unlike desktop applications used in the literature [28]–[35], [37], web applications offer significant advantages such as

the possibility to collect larger data regardless of the time and place, extension and the possibility of integration, setup simplicity and ease of use, and availability of use on multiple devices like tablets and smartphones.

A. EmoSurv: Our Web Application for the data collection

EmoSurv (www.emosurv.tech) is a dynamic web application that captures data and collects temporal and frequency features about keyboard typing dynamics while a specific emotion is induced in a user. The block diagram of the developed web application is described in Fig .5. It contains two sides: client-side (front-end) and server-side (back-end). The back-end of the application was performed using Node.js. The client-side was developed using TypeScript, HTML, and CSS under the Angular framework. It realizes the following functions:

- Creation of the user interfaces
- Creation of a dashboard for data visualization
- Generation of time and frequency features and transmitting them to the server

The collected data is stored and managed by MongoDB. To perform the data collection process, the application guides each participant to go through the following tasks (as in Fig .6):

- Type a free and a fixed text before emotion induction. We suppose that the user is in his neutral state.
- Watch an emotion-eliciting video. Videos are carefully chosen based on [40].
- To ensure that the participant watched the whole video and was totally engaged, he is asked to respond to an accuracy question. Data belonging to participants who answered wrong, are discarded in the data cleaning process.
- Type another free and fixed texts just after finishing the video.
- Answer some questions about personal information like age range, sex, number of fingers used to type (one-finger-typist, two-finger-typist, and touch-typist), average time spent typing on keyboard per day, educational level, and profession. These information are used to analyze users’ performance.
- At the end, the participant can choose to leave the application, or continue with another video to experience another emotional state.

B. Emotion model

In this research work, we opt for the discrete emotion model introduced by Paul Ekman [4] which consists of 6 emotions: anger, disgust, fear, happiness, sadness, and surprise.

C. Feature extraction

The raw keystroke data consists of the key code, the timestamp of key press (Key down) and key release (key up). To generate features, we used digraphs (a combination of 2 consecutive keystroke events) and trigraphs (a

TABLE III: Accuracy rates and techniques used for classification

Approaches	Techniques used	Results accuracy	Evaluation measure	Ref.
Machine learning	ANN and SVM	91.24% accuracy and 4.35% FP rate using SVM (fright)	Accuracy / false positive and false negative	[29]
	C4.5 and Decision tree	77% to 88% for hesitance, nervousness, relaxation, sadness, and tiredness	10-fold cross validation / true positive / false positive / Kappa	[30]
	KNN, KStar, Random Committee, Random Forest, and Bounded K-means Clustering	69% using RandomComittee for positive and negative emotions	false positive and false negative / Accuracy	[31]
	decision tree, neural networks, K nearest neighbors, naive bayes, AdaBoost, Rotation forest, and Bayesian network	81.25% using AdaBoost for fear	Recall	[32]
	C4.5, tree J48, Random Forest, SVM, and Naive Bayes	80,6% for negative and positive arousal using Random Forest	Accuracy / Kappa / Ref. acc / Acc Imp	[28]
	Simple logistics, SMO, multilayer perceptron, Random Tree, J48, and BF tree	Between 70% and 80%for fixed text / 60% to 82% for free text	Success rate	[34]
	Logistic regression, SVM, Nearest neighbor, C4.5, and Random Forest	84% for arousal using KNN and 83% for valence using KNN	F-score	[36]
Statistical	2-way 3(Valence: negative, neutral, positive) * 3(Arousal: negative, neutral, positive) ANOVA	Significant ($p < 0.001$)	Accuracy	[33]
	linear regression model	Significant ($P < 0.01$)	Correlation analysis	[35]
	Logistic regression	60% to 88% accuracy with mean 72%	Nagelkerke R^2 and χ^2 / CA / FPR / FNR	[37]

TABLE IV: Different datasets constructed by retained studies

Emotion model	Number of users	Raw Data Recording methodology	Data labeling methodology	Emotion induction method	Task for data collection	Ref.
Paul Ekman's basic emotions reduced to 4	50	Desktop application following Experience Sampling Methodology (ESM)	Self-reports (the user describes his current emotion)	-	-	[29]
15 discrete emotional states	12	C# Desktop application following Experience Sampling Methodology (ESM)	Self-reports using questionnaires (3-point Likert Scale)	-	free and fixed text	[30]
Valence-Arousal-Dominance (happiness, inspiration, sympathy, disgust, fear)	35	Desktop application	Self-report using questionnaires	video clips, pictures	free text	[31]
7 discrete emotions (fear, anger, sadness, happiness, disgust, boredom, surprise)	9	Desktop application	Self-reports using questionnaires and free text box	-	Free text	[32]
2-dimensional model (Valence & Arousal)	27 and 14	Desktop application (MOKEETO)	Self-labeling (scale from 1 to 9)	Task difficulty	Free text	[28]
2-dimensional model (Valence & Arousal)	27	Desktop application developed using C#	Self-Assessment Manikin scale (9 point rating scale)	International Affective Picture System	fixed text	[33]
7 Discrete emotions (Anger, disgust, fear, joy, sadness, guilt and shame)	25	Desktop Java-based software for fixed text and C# based software for free text	Self-report using multiple answer question	-	Free and fixed text	[34]
3-dimensional model (Valence & arousal & control)	152	Application and keyboard layout for Android OS (Touchscreen)	Self-labeling using questionnaires	3-time limits	fixed text	[35]
14 emotions in a 2-dimensional model (Valence & Arousal).	64	Javascript web-based application	Self-report using 2-dimensional Figure / Think Aloud Protocol	movie clips/task	-	[36]
Stress	9	-	-	3-back delayed recall tasks	Free text	[37]

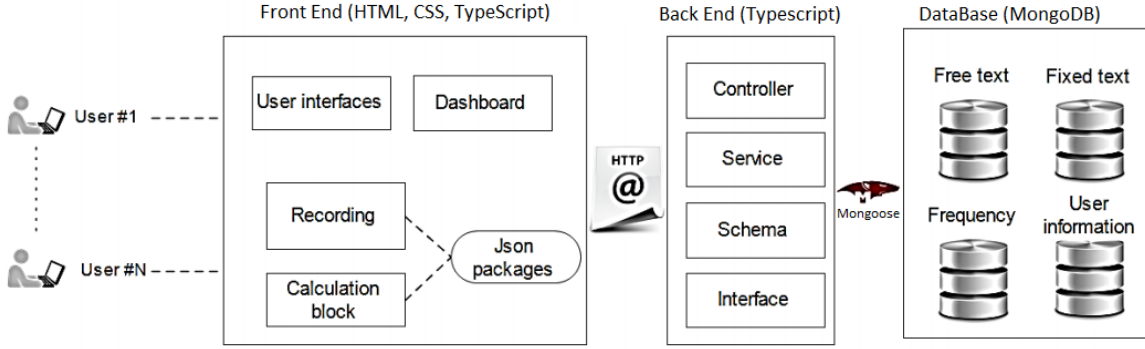


Fig. 5: Block diagram of the web application

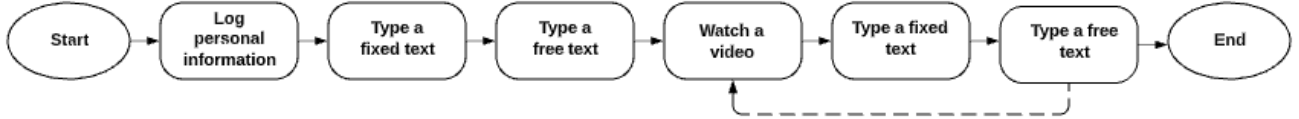


Fig. 6: Scenario of the front-end user's web-application. Dashed lines represent the case where a user chooses to continue with another video

combinations of 3 consecutive keystroke events). Table V presents the coded keystroke features used and their description. All the features were calculated for the fixed and free text, before and after the emotion elicitation.

TABLE V: Extracted features

Features	Description
D1U1	Time between first key down and first key up
D1U2	Time between first key down and second key up
D1D2	Time between first key down and second key down
U1D2	Time between first key up and second key down
U1U2	Time between first key up and second key up
D1U3	Time between first key down and third key up
D1D3	Time between first key down and third key down
Typing speed	Number of characters per unit of time
DelFreq	Relative frequency of delete key
LeftFreq	Relative frequency of backspace key

D. Preliminary findings

The survey application for dataset construction was launched on 14 January 2020. As we started to investigate the data that we have already received, we notice that, statistically, the values of features extracted are very comparable when a user goes from a neutral state to another induced emotion, or from an emotional state to another. This confirms the fact that keystroke dynamics is influenced by emotions.

VI. CONCLUSION

The main goal of this paper was to perform a systematic literature review to summarize key information in the area of emotion recognition through keystroke dynamics. To this end, we answered questions about the advantages

and disadvantages of using keystroke dynamics, keystroke features, classification algorithms applied, measures for performance evaluation, and datasets benchmarking.

Although previous works presented high accuracy results, we should note that most of them used small datasets with a low number of users. Users' typing skills, educational level, the influence of age, sex, and many other parameters were not taken into account when classifying emotions in most of the works. Therefore, more datasets that take into consideration these parameters and assign keystroke patterns to emotion are needed in order to increase performance and encourage research in this field.

In addition to summarizing and discussing key information in the area of emotion detection through keystroke dynamics, this paper describes steps for building a new dataset. As near future work, the constructed dataset will be available to researchers and will be used either for feature selection purposes by using association rule mining, or for classification by applying machine learning algorithms.

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